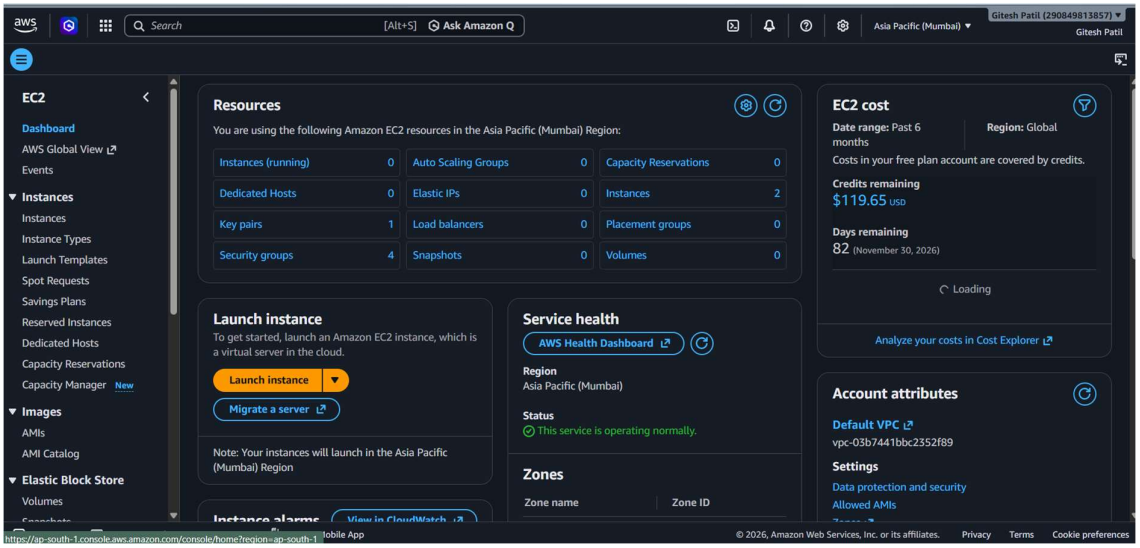
# Lab 03: Launch EC2 Instance with User Data

In this lab, we will launch an EC2 instance and use User Data to automate initial configuration tasks such as installing packages, configuring services, or running scripts at boot.

Step 1: Launch a Bastion Host

1. Navigate to EC2 Console -> Launch Instance




2. Configure :

Name: EC2 User Data

AMI: Amazon Linux

Instance type: t3.micro

3. In the Key pair (login) section click on Create new key pair and on the subsequent screen configure as follows:

Name: TestPair

Key pair type: RSA

Private key file format: .ppk

After entering these details click on Create key pair button and save the key pair file securely on your local machine.

4. Now, click on the Edit button next to Network settings and configure as follows:

VPC : Custom VPC

Subnet : Select any public subnet

Auto assign Public IP : Enable

Security group name : EC2 SG

For the Inbound rule :

1. Type : SSH

Source type : Anywhere

2. Type : HTTP

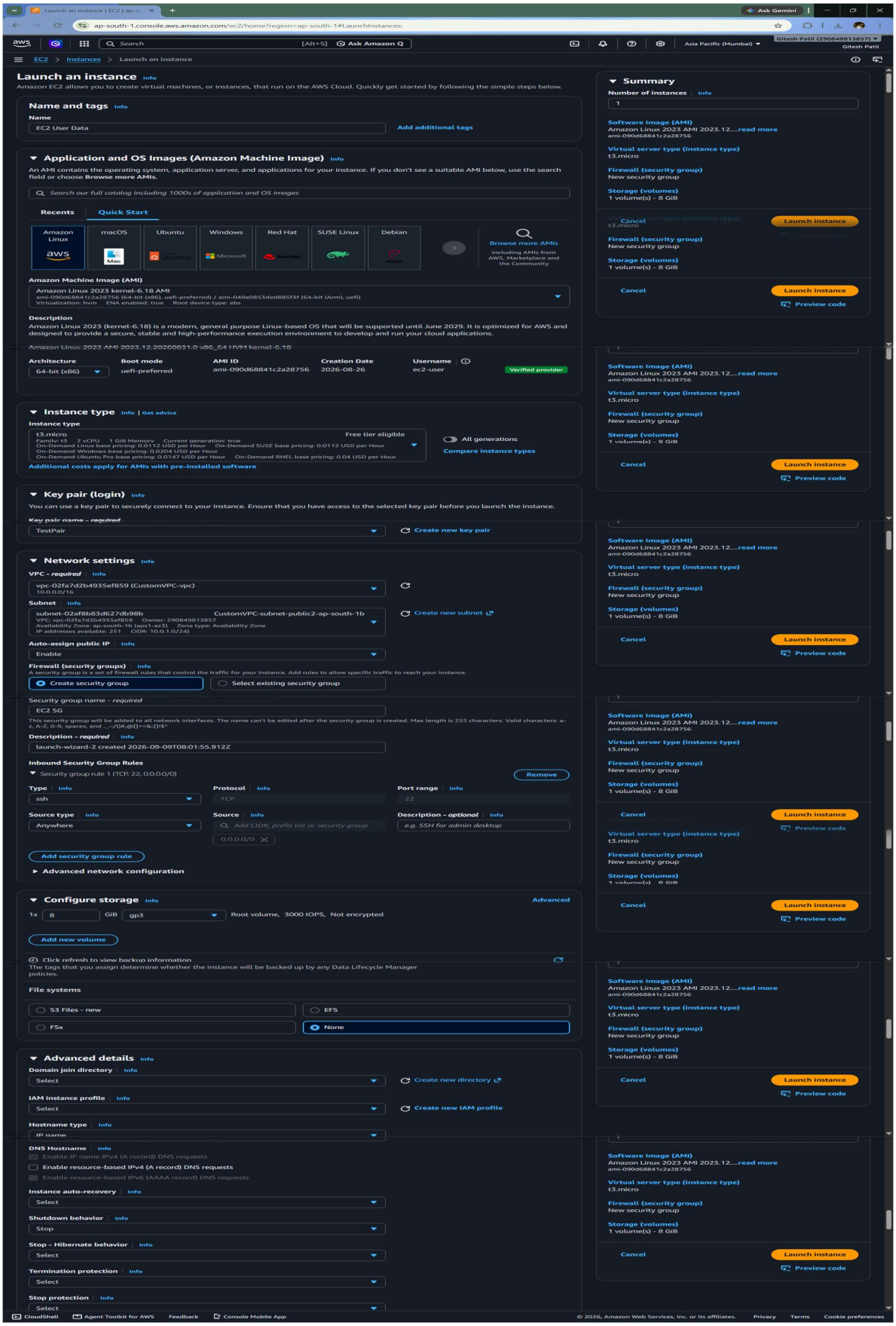
Source type: Anywhere

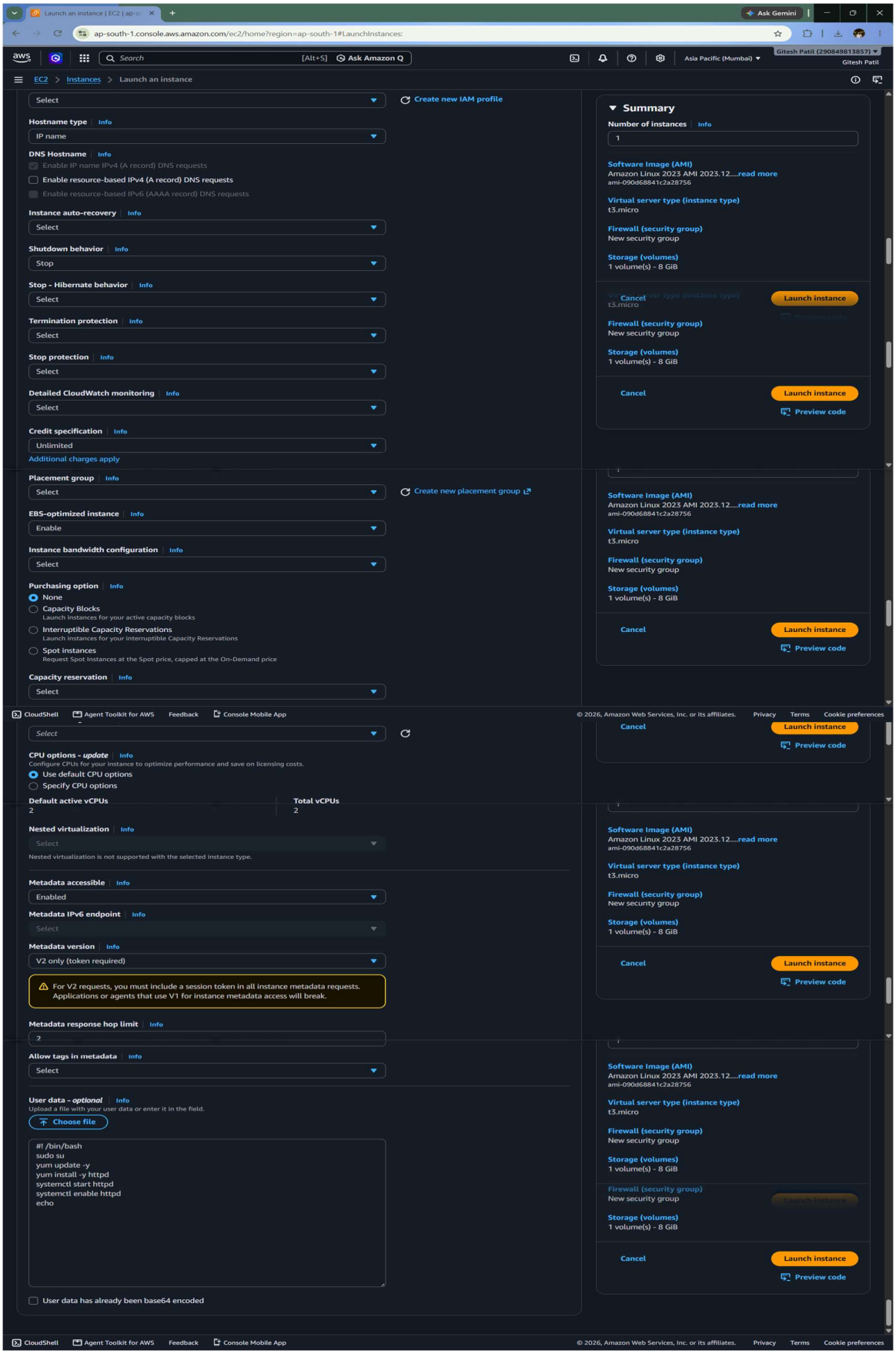
5. Expand the Advanced details section and scroll down to the User Data section and

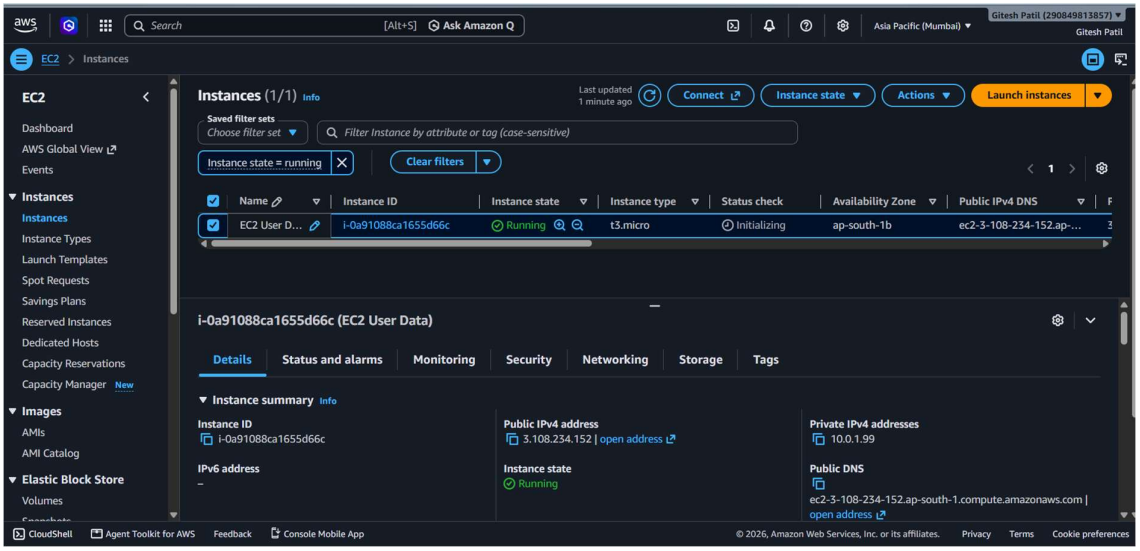
paste the following:

```
#! /bin/bash
sudo su
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
echo
```

6. After configuring all this, just click on Launch instance button and get to the instance details page.


Launch an instance | EC2 | ap-s
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:
Search [Alt+S] Ask Amazon Q
Asia Pacific (Mumbai)
Gitesh Patil (290849813857)
EC2 > Instances > Launch an instance
Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
EC2 User Data
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.18 AMI
ami-090d68841c2a28756 (64-bit (x86), uefi-preferred) / ami-049e0853ded885f3f (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs
Description
Amazon Linux 2023 (kernel-6.18) is a modern, general purpose Linux-based OS that will be supported until June 2029. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.
Amazon Linux 2023 AMI 2023.12.20260831.0 x86_64 HVM kernel-6.18
Architecture
64-bit (x86)
Boot mode
uefi-preferred
AMI ID
ami-090d68841c2a28756
Creation Date
2026-08-26
Username
ec2-user
Verified provider
Instance type Info | Get advice
Instance type
t3.micro
Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0112 USD per Hour On-Demand SUSE base pricing: 0.0112 USD per Hour
On-Demand Windows base pricing: 0.0204 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour On-Demand RHEL base pricing: 0.04 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
TestPair
Create new key pair
Network settings Info
VPC - required Info
vpc-02fa7d2b4935ef859 (CustomVPC-vpc)
10.0.0.0/16
Subnet Info
subnet-02af8b83d627db98b CustomVPC-subnet-public2-ap-south-1b
VPC: vpc-02fa7d2b4935ef859 Owner: 290849813857
Availability Zone: ap-south-1b (apn1-az3) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.1.0/24)
Create new subnet
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Security group name - required
EC2 SG
This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*
Description - required Info
launch-wizard-2 created 2026-09-09T08:01:55.912Z
Inbound Security Group Rules
Security group rule 1 (TCP, 22, 0.0.0.0/0)
Remove
Type Info
ssh
Protocol Info
TCP
Port range Info
22
Source type Info
Anywhere
Source Info
Add CIDR, prefix list or security group
0.0.0.0/0
Description - optional Info
e.g. SSH for admin desktop
Add security group rule
Advanced network configuration
Configure storage Info
Advanced
1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted
Add new volume
Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.
File systems
S3 Files - new
EFS
FSx
None
Advanced details Info
Domain join directory Info
Select
Create new directory
IAM instance profile Info
Select
Create new IAM profile
Hostname type Info
IP name
DNS Hostname Info
Enable IP name IPv4 (A record) DNS requests
Enable resource-based IPv4 (A record) DNS requests
Enable resource-based IPv6 (AAAA record) DNS requests
Instance auto-recovery Info
Select
Shutdown behavior Info
Stop
Stop - Hibernate behavior Info
Select
Termination protection Info
Select
Stop protection Info
Select
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-090d68841c2a28756
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Launch an instance | EC2 | ap-s
Ask Gemini
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:
Search [Alt+S] Ask Amazon Q
Asia Pacific (Mumbai)
Gitesh Patil (290849813857)
Gitesh Patil
EC2 > Instances > Launch an instance
Select
Create new IAM profile
Hostname type Info
IP name
DNS Hostname Info
Enable IP name IPv4 (A record) DNS requests
Enable resource-based IPv4 (A record) DNS requests
Enable resource-based IPv6 (AAAA record) DNS requests
Instance auto-recovery Info
Select
Shutdown behavior Info
Stop
Stop - Hibernate behavior Info
Select
Termination protection Info
Select
Stop protection Info
Select
Detailed CloudWatch monitoring Info
Select
Credit specification Info
Unlimited
Additional charges apply
Placement group Info
Select
Create new placement group
EBS-optimized instance Info
Enable
Instance bandwidth configuration Info
Select
Purchasing option Info
None
Capacity Blocks
Launch instances for your active capacity blocks
Interruptible Capacity Reservations
Launch instances for your interruptible Capacity Reservations
Spot instances
Request Spot Instances at the Spot price, capped at the On-Demand price
Capacity reservation Info
Select
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-090d68841c2a28756
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Select
CPU options - update Info
Configure CPUs for your instance to optimize performance and save on licensing costs.
Use default CPU options
Specify CPU options
Default active vCPUs
2
Total vCPUs
2
Nested virtualization Info
Select
Nested virtualization is not supported with the selected instance type.
Metadata accessible Info
Enabled
Metadata IPv6 endpoint Info
Select
Metadata version Info
V2 only (token required)
For V2 requests, you must include a session token in all instance metadata requests. Applications or agents that use V1 for instance metadata access will break.
Metadata response hop limit Info
2
Allow tags in metadata Info
Select
User data - optional Info
Upload a file with your user data or enter it in the field.
Choose file
#! /bin/bash
sudo su
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
echo
User data has already been base64 encoded


Search
[Alt+S]
Ask Amazon Q
Asia Pacific (Mumbai)
Gitesh Patil (290849813857)
Gitesh Patil
EC2
Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager
New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Instances (1/1) Info
Last updated
1 minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Instance state = running
Clear filters
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
EC2 User D...
i-0a91088ca1655d66c
Running
t3.micro
Initializing
ap-south-1b
ec2-3-108-234-152.ap-...
i-0a91088ca1655d66c (EC2 User Data)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-0a91088ca1655d66c
IPv6 address
–
Public IPv4 address
3.108.234.152 | open address
Instance state
Running
Private IPv4 addresses
10.0.1.99
Public DNS
ec2-3-108-234-152.ap-south-1.compute.amazonaws.com | open address
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Instances | EC2 | ap-south-1
It works! Apache httpd
Ask Gemini
Not secure
3.108.234.152

It works!